

Fakiha Malik

Computer Engineer

Contact

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Technical Skills

- Programming & Development:
Python, Web Development (HTML, CSS, basic JavaScript), Basic SQL
- Tools & Software:
Google Colab, Anaconda, MATLAB, Oracle, Visual Studio
- Simulation & Networking:
Proteus, Cisco Packet Tracer
- Operating Systems:
Linux (Command Line Proficiency)
- Embedded Systems & Microcontrollers:
Arduino, ESP32, Basic AVR, Sensor Interfacing (I2C, SPI, UART)
- Hardware Lab Skills:
Breadboarding, Multimeter, Oscilloscope, DC Power Supply, LED Circuits, Resistor Color Coding, Series/Parallel Circuits

Language

English (Proficient)

Urdu (Native)

Summary

I am a Computer Engineering and Technology student with a strong foundation in hardware and software systems. Skilled in problem-solving, programming, and digital design through academic projects and practical labs. Passionate about learning new technologies and applying engineering concepts to real-world solutions.

Education

Intermediate (Pre-Engineering) 2021-2023
Islamabad Model College for Girls (IMCG) G-10/4

Bachelor of Science in Computer Engineering Technology (BSCET) 2023-Present
International Islamic University Islamabad (IIUI)

Projects

Smart Parking System

Arduino, Ultrasonic Sensor, Servo Motor, LEDs

- Built an automated parking barrier system using Arduino
- Used ultrasonic sensor to detect vehicle arrival and departure
- Controlled servo motor to open/close gate automatically
- Displayed parking slot status using LEDs

VPN Server on Rocky Linux

Rocky Linux, WireGuard, Command Line

- Configured a VPN server on Rocky Linux for secure remote access
- Managed user authentication and encryption settings
- Gained hands-on experience with Linux networking and firewall rules

Music Streaming Website

HTML, CSS, JavaScript

- Built a multi-page website (Home, About, Contact Us, Music) with seamless navigation
- Created nested artist → album → song structure: 6 artist containers, each linking to albums and playable songs
- Integrated custom audio player with playback speed control
- Applied front-end interactivity, event handling, and dynamic content rendering

Brain-Computer Interface (BCI) for Cursor Control using EEG Signals (Ongoing)

iWorx EEG Headset, Python, Machine Learning, Signal Processing

- Acquired real-time EEG signals using iWorx headset to capture blink patterns (single blink, double blink, etc.)
- Preprocessed and extracted features from raw EEG data for blink detection
- Trained a machine learning model to classify different blink types
- Mapped classified blinks to cursor control commands on screen
- Aiming to enable hands-free cursor navigation using intentional blinks